Mathematics

Year 5

Cambridge Primary

Student Manual

About this book

Mathematics

YEAR 5 · STUDENT BOOK · PRIME SCHOOL

The Measuring Table

This book is a table with things on it: a terracotta disc, a sage triangle, an ochre bar, a navy cylinder, a wooden ruler, a pair of compasses. Somebody has laid them out in the light and is about to measure them.

Year 5 is the year that happens to numbers. Until now a number could usually be counted. From here it is measured, which means it lands between two whole numbers and needs a way of saying exactly where. That way is a decimal, and Unit 1 is about nothing else.

Eighteen units follow the Prime School Year 5 syllabus, 150 hours across three terms, each laid out on the same table: looked at, taken apart, written down, and checked a second way.

IMPRINT

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The words, examples, numbers, diagrams and photographs are original to this edition. Every factual claim is sourced at the back, and every answer was computed rather than remembered.

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Welcome

You already measure things

You know how tall you are, and you did not say it in whole metres.

Somewhere in the answer there was a point, and after the point there were digits that mattered: one metre and forty-two centimetres is 1.42 m, and the 4 and the 2 are the reason the number is any use. A doctor who wrote down 1 m would have measured nothing.

This is the year those digits get names, and the names are the whole idea. The 4 in 1.42 is not four. It is four tenths of a metre, which is forty centimetres, which you can hold your hands apart and show. The 2 is two hundredths, which is two centimetres, about the width of two fingers. A digit is worth what its place says it is worth, and the place after the point carries on exactly as the places before it do: each step to the right is ten times smaller.

Once that is clear, three things stop being tricks and become obvious. Why multiplying by ten shifts every digit one place to the left. Why 0.7 is bigger than 0.07 even though seven is seven. Why a number with a point can be rounded to a whole one, and why the rule for doing it is an agreement rather than a fact.

One instruction for the year

Write the unit. Not 75.6, but 75.6 metres. A number without its unit is the commonest wrong answer in this subject, and it is wrong in a way that is hard to notice, because it looks finished.

Where to be careful

In Portugal you write a comma where this book writes a point: 1,42 m on a sign in Lisbon is 1.42 m in these pages. Both mean the same measurement. Cambridge mathematics uses the point, so this book uses the point, and every time you copy a number off a Portuguese label you will need to swap the mark. Doing that swap carefully is the first real skill of the year.

How to use this book

Every unit is laid out the same way, so you learn the furniture once.

On the table today	A question from real life, before any teaching.
Do you remember?	Something from an earlier year that this page leans on.
How it works	The teaching. Short on purpose: if it looks short, read it twice.
Watch me lay it out	One example done slowly, including the step people skip.
Check it another way	The same answer by another route. One method is a trick, two is understanding.
Your turn	Numbered questions with room to write. The answers are at the back, and reading them first is the one way to waste them.
One step further	For whoever finishes early, and anybody curious.
From the notebook	A real fact, with a real source at the back.
Watch the units	A warning where the unit is the trap, not the arithmetic.
Table challenge	A harder question with more than one answer.
Beyond the page	A page worth your time, with the address written out as well.
Along the way	The unit project. Longer than a lesson, and the part you remember.
What you can do now	What the unit taught, in the order it taught it.
How sure are you?	Your own honest mark against each. Nobody else fills this in.

THE FURNITURE OF A PAGE

- The **strand colour** on the outer edge: terracotta for number, sage for geometry and measure, navy for statistics, ochre for the laws of arithmetic, brass for projects.
- The **running head** says which unit and topic you are in, and the page number sits in the coloured tab in the outer corner.
- A rule is a line to write on, a box is space for working. If a question needs more room than the book gives it, use your notebook.

Your toolkit

PLATE 1 The table, set for Unit 1: a rule, a tape, a pencil, an eraser, blank cards, and a dish of blank discs standing in for coins. Coins run through this whole unit, because their masses and widths are decimals somebody else measured and you can check.

WHAT YOU NEED THIS YEAR

- A pencil, a rubber, and a 30 centimetre ruler with millimetres on it.
- A notebook with squares rather than lines. Decimals need columns.
- Digit cards from 0 to 9. Make them from card, one digit per card.
- A metre rule for the class, and a tape measure if there is one.
- A small handful of euro coins, or paper ones cut to the right size.
- A calculator, used only where a page says so, because there the arithmetic is not the point of the question.

HOW THE TABLE RUNS

1. **Measure twice.** A measurement taken once is a guess with a number on it.
2. **Write the unit.** Every time. Metres, grams, euros, degrees.
3. **Line up the columns.** Ones under ones, tenths under tenths. Most decimal mistakes are alignment mistakes, not arithmetic ones.
4. **Check it another way.** If the second method disagrees with the first, you have found something out. That is a good position to be in.
5. **Say the number out loud**, digit by digit after the point. What you say is what you understand.

1

The number system

Decimals, place value, and rounding

IN THIS UNIT YOU WILL

- √ Explain what a digit is worth in a decimal number, down to hundredths.
- √ Multiply and divide whole numbers by 1000.
- √ Multiply a decimal by 10 and by 100.
- √ Round a number with one decimal place to the nearest whole number.

1.1 Understanding place value

ON THE TABLE TODAY

Two pupils measure the same classroom. One writes **8 m**. The other writes **8.40 m**.

Both used the same metre rule and neither made a mistake. Only one of them can tell you where to cut the carpet. Which one, and what did the other throw away?

DO YOU REMEMBER?

In a whole number, each place to the **left** is ten times bigger: ones, tens, hundreds, thousands. Nothing about that pattern stops at the ones column.

HOW IT WORKS

The places carry on to the right of the ones in the same way: ten times smaller at every step. A point after the ones shows where the whole numbers end.

H HUNDREDS	T TENS	O ONES		T TENTHS	H HUNDREDTHS	TH THOUSANDTHS
	2	3	.	2	5	
	20	3		0.2	0.05	

A one euro coin measures 23.25 millimetres across.

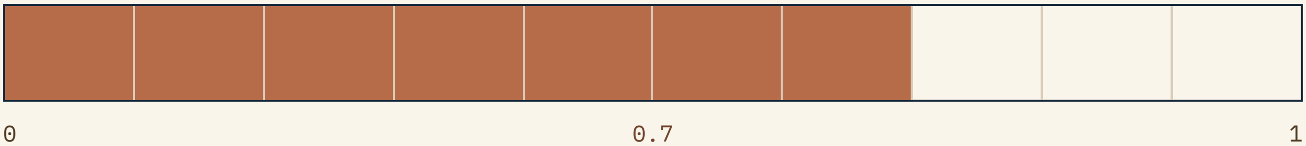
Read the bottom row: the 2 on the left is worth **20**, the 3 is worth 3, the 2 after the point is **2 tenths** and the 5 is **5 hundredths**. Add them up and the number comes back:

$$20 + 3 + 0.2 + 0.05 = 23.25$$

The same digit 2 appears twice and is worth 20 in one place and 0.2 in the other. That is the whole idea, in one measurement. The digits after the point are not decoration and not a smaller second number: a coin machine that ignored them would take the wrong coin.

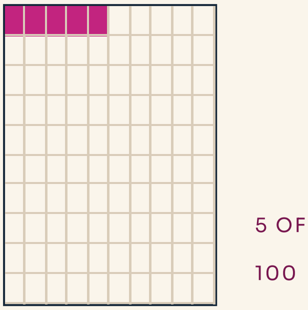
HOW IT WORKS, CONTINUED

A tenth is what you get when one whole is cut into ten equal parts. Ten tenths make one.



0.7 of the strip is shaded. Seven tenths.

A hundredth is each of those tenths cut into ten again. A hundred hundredths make one, and ten of them make one tenth.



0.05 of the square is shaded. Five hundredths, which is five of the hundred small squares, and exactly half of one tenth.

This is why **0.7 is larger than 0.07**, although seven is seven in both. Seven tenths is most of a whole. Seven hundredths is not quite a tenth of a whole. The digit tells you how many. The place tells you how big.

WATCH THE UNITS

The 5 in 23.25 millimetres is five hundredths of a **millimetre**, not five millimetres. When you say a digit's value out loud, say the unit with it, and the mistake becomes impossible to make.

FROM THE NOTEBOOK

The measurements in this unit come from the European law that says what a euro coin must be: Council Regulation (EU) No 729/2014. Its own table prints the diameter of a one euro coin as **23,25 mm**, with a comma, because that is the mark most of Europe writes. This book prints 23.25 mm. Same coin, different mark.

Say it as it is, too: twenty-three point two five, digit by digit after the point. "Point twenty-five" turns two places into one number that does not exist.

WATCH ME LAY IT OUT

What is each digit worth in 19.75? A 10 cent coin measures 19.75 millimetres across.

1 Write it into the chart, one digit per column, points lined up.

H HUNDREDS	T TENS	O ONES		T TENTHS	H HUNDREDTHS	TH THOUSANDTHS
	1 10	9 9	.	7 0.7	5 0.05	

2 Name each digit by its column, not by its look.

DIGIT	COLUMN	WORTH
1	tens	10
9	ones	9
7	tenths	0.7
5	hundredths	0.05

3 Say it out loud: one ten, nine ones, seven tenths, five hundredths.

Adding those four values gives 19.75 back, which is the check. Notice the sizes going left to right: 10, 9, 0.7, 0.05, each a tenth of the one before it.

CHECK IT ANOTHER WAY

Break the number apart and add it back. If the parts do not rebuild it, one is in the wrong column.

$$10 + 9 + 0.7 + 0.05 = 19.75$$

Then check it against something you can hold. A 10 cent coin is 19.75 mm across and a 5 cent coin is 21.25 mm, so the 5 cent coin is wider although it is worth less. 21 ones beats 19 ones, so start every comparison at the biggest place.

YOUR TURN

Work in your notebook or in the spaces given. Line up your columns.

1 What is each digit worth? a. 21.25 b. 8.5 c. 2.3

2 Write each number as an addition of its places, as in the worked example. a. 25.75 b. 7.5 c. 0.06

3 Write the number made from: a. 2 ones, 9 tenths and 5 hundredths b. 7 ones and 8 tenths c. 3 tenths and 3 hundredths

YOUR TURN, CONTINUED

4 Which is larger? Explain how you know, using the word **place**. a. 0.7 or 0.07 b. 0.5 or 0.45 c. 2.3 or 2.30

Then write one sentence saying which pair was the odd one out, and why.

5 These are the masses of six euro coins, in grams, as set out in European law. Put them in order, lightest first.

COIN	MASS IN GRAMS	YOUR ORDER
1 cent	2.3	
5 cent	3.9	
10 cent	4.1	
20 cent	5.7	
50 cent	7.8	
1 euro	7.5	

HOW IT WORKS

Multiplying by ten does not move the point. **The point never moves.** The digits move, one place to the left, because every one of them is now worth ten times what it was worth.

7.5   7 . 5

× 10   EVERY DIGIT MOVES 1 PLACE TO THE LEFT

75   7 5 .

One euro coin has a mass of 7.5 g. Ten of them have a mass of 75 g.

Ten times a tenth is a whole one, so the digit that was in the tenths column arrives in the ones column. Nothing is added and nothing is invented. Each digit takes one step to the left.

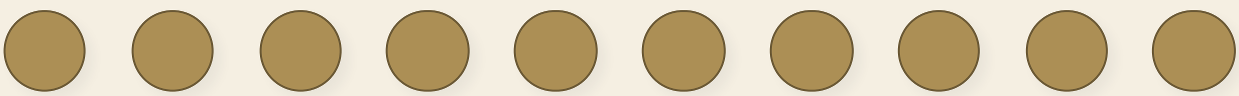
Multiplying by a hundred is the same move made twice.

7.5   7 . 5

× 100   EVERY DIGIT MOVES 2 PLACES TO THE LEFT

750   7 5 0 .

A hundred one euro coins have a mass of 750 g, which is three quarters of a kilogram.



7.5 G EACH

$10 \times 7.5 \text{ g} = 75 \text{ g}$

Ten of them. Ten times something is easier to believe when you can count the ten, so this row is drawn by the build rather than photographed: a picture that carries a number has to be right.

WATCH THE UNITS

3.9 g × 10 = 39 g, and **not** 39 kg. Multiplying the number does nothing at all to the unit. If the unit changes in your answer, you have changed the question.

HOW IT WORKS

A thousand is ten, ten times, and then ten again. So multiplying a whole number by 1000 moves every digit **three** places to the left, and dividing by 1000 moves every digit three places to the right.

76	7 6
	× 1000 EVERY DIGIT MOVES 3 PLACES TO THE LEFT
76000	7 6 0 0 0

1993	1 9 9 3 .
	÷ 1000 EVERY DIGIT MOVES 3 PLACES TO THE RIGHT
1.993	1 . 9 9 3

The highest point of mainland Portugal is the Torre in the Serra da Estrela, at 1993 metres. In kilometres that is 1.993 km, because a thousand metres make one kilometre.

Look at what dividing by 1000 did to the digits of 1993. The 1 was worth a thousand and is now worth one. The 9 that was worth ninety is now worth nine hundredths. Every digit went three places to the right and the point stayed where it always is, after the ones.

That is also why ÷ **1000 turns metres into kilometres** and × **1000 turns kilometres into metres**. The two operations are the same fact read in opposite directions.

CHECK IT ANOTHER WAY

Check by size, before you check by digits. 1993 metres is nearly two thousand metres, so in kilometres it must be nearly two. 1.993 is nearly two. If your answer had been 19.93 or 0.1993, that first check would have caught it in a second, and it costs nothing.

YOUR TURN

6 Multiply by 10. a. 4.1 b. 2.3 c. 3.9

7 Multiply by 100. a. 7.5 b. 8.5 c. 0.06

8 Multiply by 1000. a. 76 b. 45 c. 8

9 Divide by 1000. a. 1993 b. 12 500 c. 750

10 A 50 cent coin has a mass of 7.8 g. What is the mass of a roll of 20 of them? Give the answer in grams and in kilograms, and show your working.

11 A rope is 8.40 metres long. Write that length in centimetres. Which operation did you use, and why?

ONE STEP FURTHER

The Vasco da Gama bridge crosses the Tejo north of Lisbon. Its central viaduct is 6.351 km long and its south viaduct is 3.825 km long.

1. In 3.825 km, what is the 8 worth? Give your answer in kilometres and then in metres.
2. In 6.351 km, what is the 5 worth? What is the 1 worth?
3. Write both lengths in metres. Which operation did you use?
4. The whole crossing, road included, is 17.2 km. Is that longer or shorter than the two viaducts added together? Show how you decided.

A third place after the point is thousandths, and it behaves exactly like the two places before it: ten times smaller again. Nothing new has to be learnt to read it, which is the whole point of a place-value system.

BEYOND THE PAGE

Place Value, ages 7 to 11

nrich.maths.org/curriculum/place-value-age-7-11

Twenty-five free place-value activities from NRICH, at the University of Cambridge. Start with a puzzle rather than an explanation: if you can win the game, you understood the place.

1.2 Rounding decimal numbers

ON THE TABLE TODAY

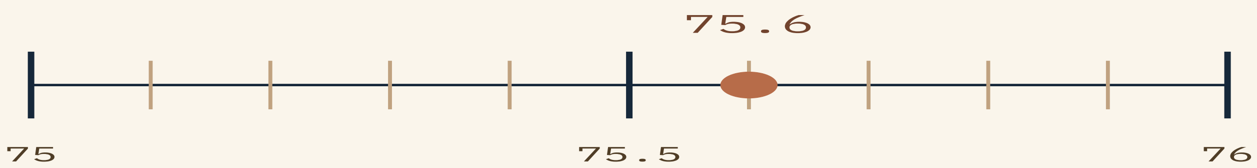
The Torre dos Clérigos in Porto is **75.6 metres** tall. A poster in the ticket office says **76 metres.**

Somebody has changed the number. Is the poster wrong, and would you print it?

HOW IT WORKS

Rounding a decimal to the nearest whole number is one question asked properly: **which whole number is it closer to?**

Put the number on a line between the two whole numbers on either side of it, and look.



THE HALFWAY MARK IS 75.5

75.6 sits past the halfway mark, so it is closer to 76 than to 75.

The halfway mark between 75 and 76 is 75.5. Anything above it is nearer 76. Anything below it is nearer 75. That is the whole of rounding, and the rule schools teach is only a shortcut for reading this line without drawing it:

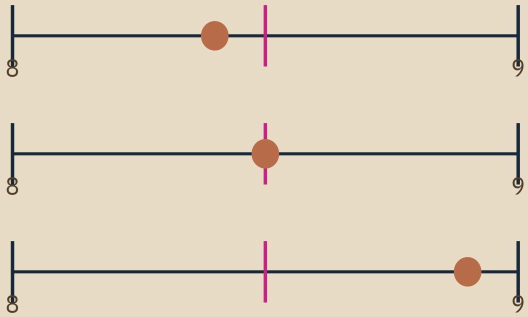
Look at the tenths digit. If it is 5 or more, round up. If it is 4 or less, round down.

The poster is not wrong. It is 75.6 metres rounded to the nearest metre, and for a poster that is the right thing to print. For a builder ordering the stone it would not be.

Three numbers exist for this one tower: an official tourism page says the tower rises through **75 metres**, a guide gives **75.6 metres**, and the poster says **76 metres**. Only one of the three is a measurement. The other two are that measurement rounded, in two different directions, by two people who each had a different reason.

WATCH ME LAY IT OUT

Round 8.4, 8.5 and 8.9 to the nearest whole number.



8.4	NEARER 8	8
8.5	ON THE HALFWAY MARK	9
8.9	NEARER 9	9

1 **8.4** The tenths digit is 4, which is less than 5, so 8.4 is nearer 8. It rounds to **8**.

2 **8.9** The tenths digit is 9, which is more than 5, so 8.9 is nearer 9. It rounds to **9**.

3 **8.5** The tenths digit is exactly 5. Look at the line: 8.5 is not nearer to either one. It is exactly halfway.

So 8.5 is the interesting one. Nothing in the mathematics decides it, because there is nothing to decide: the two distances are equal. What decides it is an **agreement**, and the agreement is to round up. 8.5 rounds to **9**.

An agreement is not a weaker kind of rule. It is what makes two people who round the same number get the same answer, which is the only reason rounding is any use in the first place.

CHECK IT ANOTHER WAY

Cover the tenths digit with your finger and read the whole number that is left. That is the number below. Add one to it and you have the number above. Now ask yourself only one question: is the part I covered up more or less than half?

It is the same decision as the line, made without drawing anything, and it works just as well for 249.7 as it does for 8.4.

YOUR TURN

12 Round each of these to the nearest whole number. a. 12.3 b. 0.9 c. 7.5 d. 3.1 e. 49.6 f. 99.5

13 Each of these is a real measurement. Round it as the question asks, and keep the unit in your answer. a. The Torre dos Clérigos is 75.6 m tall. To the nearest metre. b. A 20 cent coin has a mass of 5.7 g. To the nearest gram. c. The Torre in the Serra da Estrela is 1.993 km high. To the nearest kilometre.

14 A number with one decimal place rounds to 14. Write down three numbers it could be, and one number it could not be.

15 Marta says: "0.4 rounds to 0, so 0.4 is nothing." Explain, in one sentence, what is wrong with the second half of that sentence.

ONE STEP FURTHER

Round at the end, not at the start. Two lengths of ribbon measure 2.5 m each.

- Add them first, then round: 2.5 + 2.5 = 5, and 5 is already whole. **5 m.**
- Round them first, then add: 3 + 3 = **6 m.**

A whole metre of ribbon has appeared out of nothing, because the rounding happened too early. Rounding always loses a little, and rounding before you calculate loses it in the middle of the working, where it grows.

1. Find two numbers with one decimal place where rounding first gives an answer one less than rounding last.
2. A shop adds up 8 items priced 1.5 euros each. Work out the true total, then the total if every price is rounded first. Which is bigger, and by how much?

FROM THE NOTEBOOK

A one euro coin has a mass of 7.5 g. A fifty cent coin has a mass of 7.8 g.

The coin worth twice as much is the lighter of the two, so mass is not value, and the tenths digit is the only thing separating them. Check it yourself on a kitchen scale that reads tenths of a gram.

BEYOND THE PAGE

Round the Dice Decimals 1

nrich.maths.org/problems/round-dice-decimals-1

Roll two dice, make a number with one decimal place, round it. Then the real question: can two different rolls round to the same whole number? Free, from NRICH.

SIX CITIES, SIX POPULATIONS

PLATE 3 Six blocks, six heights, one table. A population is a measurement like any other, and it is far too big to count twice.

A population is a number nobody can check by counting. It is measured once every ten years by a census and then quoted by everybody. Portugal's last census was the **Censos 2021**, run by the Instituto Nacional de Estatística, and its figures are published at **censos.ine.pt**.

WHAT TO DO

1. Choose **six** Portuguese municipalities. Include the one you live in, one you have visited, and one you have only heard of.
2. Find the resident population of each in the Censos 2021, at censos.ine.pt. Write down the figure and the page you found it on.
3. Write each population **in figures**, grouped in threes with a thin gap, and then **in words**.
4. Round each one to the nearest **thousand**, and again to the nearest **ten thousand**.

YOUR SIX, BEFORE YOU START

Draft them here in pencil, so the sheet you hand in is not where you think.

SIX CITIES, SIX POPULATIONS, CONTINUED

5 Put your six municipalities in order, largest first.

6 Present the six on one sheet, as six bars drawn to scale, with the figures written on. Draw the bars from the rounded figures and write the exact figure beside each one, so a reader can see both.

THE QUESTION TO ANSWER IN WRITING

Which rounding belongs on a poster, and which belongs in a report? Answer in three or four sentences, using at least one of your own six numbers as the example.

HOW IT WILL BE JUDGED

WHAT IS BEING JUDGED	WEIGHT
Six real figures, each with the source page named	30%
Figures and words both correct, and grouped correctly	25%
Rounding correct to both places	25%
The written answer about posters and reports	20%

WHAT YOU CAN DO NOW

- √ Explain what any digit in a decimal number is worth, down to hundredths.
- √ Write a decimal as the sum of its places, and rebuild it from those places.
- √ Say why 0.7 is larger than 0.07, using the word place.
- √ Multiply a decimal by 10 and by 100, and say what happened to the digits.
- √ Multiply and divide a whole number by 1000, and change metres to kilometres.
- √ Round a number with one decimal place to the nearest whole number.
- √ Say why 8.5 rounds to 9, and why that is an agreement rather than a fact.
- √ Explain why rounding at the start of a calculation is a mistake.

WHERE EACH ONE WAS TAUGHT

WHAT YOU CAN DO	PAGE
Everything about the value of a digit, and multiplying by 10, 100 and 1000	8 onwards
Rounding a decimal to the nearest whole number	17 onwards
The project, and what it is judged on	21
Every answer, and the working where working helps	27
Every word, defined	26

WORDS ON THE TABLE

These are the words this unit used. Cover the word list at the back of the book and say what each one means, out loud, in a full sentence. Any you cannot say is a page to go back to, not a word to look up and forget.

decimal number · decimal point · place value · tenth · hundredth · thousandth

round · nearest whole number · halfway · estimate · mass · kilometre

HOW SURE ARE YOU?

Nobody else fills this in. Mark each line honestly, then go back to the page that taught the ones you marked "not yet". Being honest here saves you the time you would otherwise lose in Unit 2, which assumes all of this.

I CAN	NOT YET	NEARLY	CONFIDENT
Explain what a digit is worth in a decimal number	○	○	○
Write a decimal as the sum of its places	○	○	○
Say why 0.7 is larger than 0.07	○	○	○
Multiply a decimal by 10 and by 100	○	○	○
Multiply and divide a whole number by 1000	○	○	○
Round a number with one decimal place to the nearest whole number	○	○	○
Explain why 8.5 rounds up	○	○	○
Explain why rounding early is a mistake	○	○	○

ONE THING I WANT TO GET BETTER AT

TABLE CHALLENGE

Three questions with more than one way in. Bring your reasoning, not only the number: a right answer with no reasoning is worth less here than a wrong answer with good reasoning, because only one of the two can be corrected.

1 I am a number with one decimal place. I round to 8. My tenths digit is odd. List every number I could be, and then say how you know your list is complete.

2 Using the digit cards **3**, **5** and **8** once each, and one decimal place, make the number closest to 40. Then make the one closest to 60. How many numbers can you make in total?

3 Two different numbers with one decimal place round to the same whole number, and the difference between them is 0.8.

a. Find a pair. b. How many pairs are there for one whole number? c. Explain why the answer to b is the same for every whole number.

Word list

Every word taught so far, in alphabetical order, with the unit that taught it.

census Unit 1
A count of every person living in a country, taken by the government, in Portugal once every ten years.

column Unit 1
A place in a written number, such as tens, ones or tenths. Lining columns up is how decimals are added safely.

decimal number Unit 1
A number written with a point in it, so that parts smaller than one can be shown.

decimal point Unit 1
The mark written straight after the ones digit, showing where the whole numbers end.

diameter Unit 1
The distance straight across a circle, through its centre.

digit Unit 1
One of the ten symbols 0 to 9 that numbers are written with.

estimate Unit 1
An answer close enough to be useful, worked out on purpose rather than arrived at by accident.

halfway Unit 1
Exactly between two numbers, where neither is nearer, and where rounding needs an agreement rather than a measurement.

hundredth Unit 1
One tenth cut into ten equal parts. A hundred hundredths make one whole, and ten of them make one tenth.

kilogram Unit 1
A thousand grams.

kilometre Unit 1
A thousand metres. Divide a length in metres by 1000 to write it in kilometres.

mass Unit 1
How much matter something contains. Measured in this unit in grams, using a balance or a scale.

millimetre Unit 1
A thousandth of a metre, and a tenth of a centimetre. The smallest mark on a school ruler.

nearest whole number Unit 1
The whole number that a decimal is closest to on the number line.

place value Unit 1
What a digit is worth because of the column it sits in. The digit says how many, the place says how big.

round Unit 1
To replace a number with the nearest simpler one, on purpose, and by a rule everybody agrees on.

tenth Unit 1
One whole cut into ten equal parts. Ten tenths make one whole.

thousandth Unit 1
One hundredth cut into ten equal parts. A thousand thousandths make one whole.

Answers

UNIT 1, TOPIC 1.1

Check the working, not only the number. Where a question asks you to explain, this is one good way of saying it, not the only one.

1 The value of each digit. a. **21.25** the 2 is worth 20, the 1 is worth 1, the 2 after the point is worth 0.2, the 5 is worth 0.05. b. **8.5** the 8 is worth 8, the 5 is worth 0.5. c. **2.3** the 2 is worth 2, the 3 is worth 0.3.

2 As an addition of places. a. 25.75 = 20 + 5 + 0.7 + 0.05 b. 7.5 = 7 + 0.5 c. 0.06 = 0.06, because six hundredths is the only place used.

3 a. 2.95 b. 7.8 c. 0.33

4 a. **0.7**, because 7 tenths is a bigger place than 7 hundredths. b. **0.5**, because 5 tenths beats 4 tenths and the 5 hundredths in 0.45 cannot catch up. c. **Equal**: the trailing zero says the hundredths column is empty. The odd one out is c, the only pair that is not really a comparison.

5 2.3 g, 3.9 g, 4.1 g, 5.7 g, 7.5 g, 7.8 g: 1 cent, 5 cent, 10 cent, 20 cent, one euro, 50 cent. The one euro coin is lighter than the 50 cent coin.

6 a. 41 b. 23 c. 39

7 a. 750 b. 850 c. 6

8 a. 76 000 b. 45 000 c. 8000

9 a. 1.993 b. 12.5 c. 0.75

10 20 × 7.8 g = **156 g**, which is 0.156 kg. Twenty lots of 7.8 is two lots of 78, and 78 doubled is 156, so no written method is needed.

11 8.40 m = **840 cm**. Multiply by 100, because one metre is a hundred centimetres, so there are a hundred times as many centimetres as metres.

ONE STEP FURTHER, THE VIADUCTS

1 The 8 is worth **0.8 km**, which is **800 m**. **2** The 5 is worth 0.05 km, which is 50 m. The 1 is worth 0.001 km, which is 1 m. **3** 3.825 km = 3825 m and 6.351 km = 6351 m. Multiply by 1000. **4** The two viaducts together are 10.176 km. The whole crossing is 17.2 km, so the crossing is longer, by **7.024 km**. The extra is the main bridge, the other viaducts and the access roads.

UNIT 1, TOPIC 1.2

12 a. 12 b. 1 c. 8 d. 3 e. 50 f. 100 For c, 7.5 is exactly halfway, so the agreement decides it. For f, 99.5 rounds to 100, and a two-digit number rounding to a three-digit one is worth noticing.

13 a. **76 m** b. **6 g** c. **2 km** Every answer keeps its unit. An answer of "76" is not finished.

14 Any three of 13.5, 13.6, 13.7, 13.8, 13.9, 14.0, 14.1, 14.2, 14.3, 14.4. It could not be 14.5, which rounds to 15, nor 13.4, which rounds to 13.

15 Rounding 0.4 to 0 does not make 0.4 into nothing. It is four tenths, and it still exists after the rounding: 0.4 + 0.4 = 0.8, which does not round to 0 at all. Rounding changes how a number is written down, not how big it is.

ONE STEP FURTHER, ROUNDING EARLY

1 For example **0.1 and 0.4**. The true sum is 0.5, which rounds to 1. Rounded first they are 0 and 0, giving 0, one less. Any pair whose tenths add up past the halfway mark while each one alone falls short will do. **2** True total: 8 × 1.5 = **12 euros**. Prices rounded first: 8 × 2 = **16 euros**, **4 euros too much**. Eight small errors, all leaning the same way.

TABLE CHALLENGE

1 7.5, 7.7, 7.9, 8.1, 8.3. Numbers that round to 8 run from 7.5 to 8.4, and five of those have an odd tenths digit. **2** The numbers you can make are 35.8, 38.5, 53.8, 58.3, 83.5 and 85.3. Closest to 40 is **38.5**, which is 1.5 away. Closest to 60 is **58.3**, which is 1.7 away. **3** For 8: **7.5 and 8.3**, or **7.6 and 8.4**. Two pairs, and only two. The numbers that round to any whole number n run from n minus 0.5 to n plus 0.4, a span of 0.9, so a gap of 0.8 fits in exactly two positions. That holds for every whole number, which is the real answer.

ALONG THE WAY, THE PROJECT

There is no answer key for the project, because the six municipalities are yours. Three things are checked instead.

- Each figure traceable: the table it came from, named, at censos.ine.pt.
- Rounding right at both places asked for, and a rounded figure never presented as the exact one.
- A written answer that commits to a position. A poster wants the nearest ten thousand, a report wants the figure as published.

Sources and references

Every fact in this book can be checked. Each line gives the claim, where it came from, and the date the page was opened.

1 Every euro coin diameter and every euro coin mass used in this unit

Council Regulation (EU) No 729/2014 on denominations and technical specifications of euro coins intended for circulation (recast), Annex I

`legislation.gov.uk/eur/2014/729/annex/I` CHECKED 2026-08-06 UNIT 1

The law that specifies the coins. Its own table prints a comma as the decimal mark, one decimal place for the masses and two for the diameters, which is why this unit uses the diameters for hundredths and the masses for rounding to whole grams.

2 The highest point of mainland Portugal is the Torre in the Serra da Estrela, at 1993 metres

Visit Portugal, the Portuguese national tourism board, page "Serra da Estrela"

`visitportugal.com/en/node/73759` CHECKED 2026-08-06 UNIT 1

The page reads: "At 1,993 metres at its highest point in Torre".

3 The Torre dos Clerigos in Porto is 75.6 metres tall and has 240 steps

Wikipedia, article "Torre dos Clerigos"

`en.wikipedia.org/wiki/Torre_dos_Clerigos` CHECKED 2026-08-06 UNIT 1

The tower's own site, torredosclerigos.pt, was checked first and publishes neither figure.

4 An official page describes the same tower as rising through 75 metres

Visit Portugal, page "Torre dos Clerigos"

`visitportugal.com/en/content/torre-dos-clerigos` CHECKED 2026-08-06 UNIT 1

Used on purpose: an official page rounds to 75, a guide gives 75.6, a poster rounds to 76. Three numbers, one tower, and only one of them a measurement.

5 The Vasco da Gama crossing is 17.2 km in total, its central viaduct is 6.351 km and its south viaduct is 3.825 km

Wikipedia, article "Vasco da Gama Bridge", infobox

`en.wikipedia.org/wiki/Vasco_da_Gama_Bridge` CHECKED 2026-08-06 UNIT 1

The 17.2 km includes the access roads, which is why it is so much longer than the two viaducts added together.

Continued. Where a source was rejected, that is recorded too: knowing why a number was left out is part of learning to use numbers.

6 Portugal's most recent census is the Censos 2021, run by the Instituto Nacional de Estatistica, published at censos.ine.pt

Instituto Nacional de Estatistica, Censos 2021

`censos.ine.pt` CHECKED 2026-08-06 UNIT 1

The project sends pupils here for their own six figures. No population number is printed in this book on purpose: provisional and definitive census results differ, so a printed figure would date and a looked-up one will not.

7 QR destination, topic 1.1

NRICH, "Place Value: Age 7-11", 25 free activities, Millennium Mathematics Project, University of Cambridge

`nrich.maths.org/curriculum/place-value-age-7-11` CHECKED 2026-08-06 UNIT 1

Free, no login needed. Decoded back from the printed code at build time, so the square really carries this address.

8 QR destination, topic 1.2

NRICH, "Round the Dice Decimals 1", ages 7 to 11

`nrich.maths.org/problems/round-dice-decimals-1` CHECKED 2026-08-06 UNIT 1

Free, no login needed. The task is the rounding question this topic teaches. Decoded back from the printed code at build time.

9 The unit plan, its numbering and its objectives

Prime School Year 5 mathematics syllabus, 150 hours, three terms, 18 units

`the syllabus document in this book's PDF/Input folder` CHECKED 2026-08-06 UNIT 1

The school's own document, and the authority for what this book must contain.

For teachers

HOW THIS UNIT IS BUILT

This is a **sample printing**: Unit 1 of eighteen, built to be looked at and marked before the rest of the book is written. The contents page shows the whole plan, and a page number printed as a dash is a unit not yet in this volume.

Syllabus mapping. Unit 1 covers the two topics of Unit 1 of the Prime School Year 5 mathematics syllabus, in the syllabus order.

SYLLABUS OBJECTIVE	WHERE IT IS TAUGHT	WHERE IT IS PRACTISED
Explain the value of a digit in a decimal number	1.1, the place-value chart and the worked example	Questions 1 to 4
Multiply a decimal by 10 and by 100	1.1, "Ten times, a hundred times"	Questions 6 and 7
Multiply and divide whole numbers by 1000	1.1, "A thousand times, and back"	Questions 8 to 11
Round a number with one decimal place to the nearest whole number	1.2, the number line and the rounding gate	Questions 12 to 15, and the challenge

What was assumed. That pupils can read and write whole numbers to at least 10 000, know that ten tenths make one, and have met tenths in Year 4. Hundredths are introduced here rather than assumed. Negative numbers are **not** used in this unit: they belong to Unit 2 and Unit 6.2 in this syllabus, and an earlier draft of this book taught them here by mistake.

Two decisions worth knowing about.

1. The masses and diameters in this unit are the legal euro coin specifications, not invented numbers. The diameters carry two decimal places and the masses one, which is why hundredths are taught with diameters and rounding with masses. A pupil can weigh the coins in class and check the book.
2. The unit says out loud that rounding a half upwards is an **agreement** rather than a mathematical fact. Pupils who ask "but why up?" are right to ask, and the honest answer is more useful than a rule stated as though it were a discovery.

WHAT TO EXPECT FROM THIS UNIT

The decimal mark. Portuguese notation uses a comma. This book uses a point, in line with the Cambridge scheme, and says so to the pupil on the Welcome page and again in Unit 1 using a real European regulation that prints 23,25 mm. Expect to keep pointing at it all year.

Answers are at the back and are computed rather than typed, by a script kept with the book. If you find an answer you disagree with, the working is reproducible, so please report it rather than correcting the page by hand.

Feedback goes in `FEEDBACK.docx` beside this book, or straight to the editor. A teacher review is a build gate here, not a suggestion, and this unit has not yet had one.